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Project Reflection

In this project, we were tasked with creating a contraption for humanitarian efforts. However, we were limited to a list of building materials and certain requirements, such as using AI to code and needing an LED, a humidity sensor, and a sensor of our choice. Given our parameters and limitations, we designed an umbrella contraption that would use the humidity sensor as its main function, intended to detect rainfall in real-world applications and spring open automatically without manual input. The LED requirement would serve as an indicator of the umbrella being used as when it's up, the LED would be on. We also decided to add a contact switch as a physical turn-off switch so that, if needed, someone can put the umbrella down with a press of a button. After deciding on our main project, we decided to give each other roles so the process could go smoothly. I was given the task of managing the breadboard, while one of my partners decided to make the physical umbrella, and the other partner decided to create the prompt that would eventually be put into ChatGPT for our code.

In our first lab, we immediately got to work as we gathered the necessary materials for the project and began working on our individual assignments so we could get as much work done as possible. On my side, I began by connecting the Arduino MicroPython chip to the board. I then connected the contact switch, LED, and resistor relatively close to each other and connected the contact switch to the Arduino using some wires. After that, I decided to connect a red and blue wire to the negative and positive sides, respectively, as they served as the connectors to the humidity sensor. I then connected the servo motor to the main Arduino using grey (ground), green (power), and yellow (signal) wires. After that, I waited for the others to wrap everything up so we could test our first version of the project. When everyone was ready, we ran the code through the Arduino. However, it seemed like the motor didn't move at all, so after rearranging the wires, we tried it again, and it seemed to work.

In our second lab, we tested our prototype yet again, and from this we learned that the PEDOT on the sensor was getting weaker and weaker. So in response to this error, my partner changed some of the code so it could adjust to the changing values. We also decided to give the LED a greater purpose in the project, so we talked about it and decided on making it a rain (humidity) intensity indicator. If humidity rose, then the LED would flash faster and faster in response to the increasing ADC and would flash less if the ADC was decreasing.

In our third lab, we tested our prototype to see if it was still working before moving on to the presentation slides. When splitting up the work, we didn't really coordinate it besides saying what slides we were doing at the current moment. However, I worked on the introduction, input and output, requirements and constraints, wiring connections, and conclusion slides while the others worked on the AI usage, flow chart, construction, physical process slides, and results slides.

Overall, the project went really well, and I believe our group as a whole has improved not only in teamwork but also in communication, as throughout the project I never really experienced any worries or

was afraid that we'd be behind, which can probably say a lot about what I think about my partners. As for the problems we've experienced, I'd say they were very minor, with the only issues I kind of found frustrating being the need to constantly change the code in response to the PEDOT on the humidity sensor weakening and, as a result, the ADC values would act erratically. Besides that, there weren't really errors that I was having a problem with.

As for what I learned from this project, I pretty much improved my breadboard skills by a pretty decent amount and learned how to adapt to ever-changing constants, such as the humidity sensor in this case. I also learned how to coordinate and evenly distribute work between people, as well as improve my communication skills by giving my teammates constant feedback and updates on my progress throughout the labs. The presentation taught me how to project my voice as a speaker, simplify my slides so they aren't too descriptive, and speak in front of a crowd. By preparing beforehand, I was able to keep my slides simple and include only the essential information, leaving the details for me to explain verbally. Overall, the project was a huge success, and it was probably the most fun part of this module, as it allowed us to get hands-on experience, use the skills we've obtained throughout the module, and allowed us to do something that we picked ourselves.